

SUBRAT ACHARYA

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Summary

Software Engineer with extensive experience in developing and enhancing software solutions for manufacturing and health technology. Proven adaptability, reliability, and a strong willingness to learn, ensuring proficiency in any task when given the opportunity. Demonstrated ability to work independently as well as in a team setting.

Education

Bachelor of Science and Engineering in Biomedical Engineering

Aug 2018 to May 2022

University of Iowa, Iowa City, IA

GPA: 4.0/4.0

Professional Experience

Software Engineer at Avantor

Aug 2022 to Present

Remote

- Architected and delivered the End of Line (EOL) tester web application for Masterflex touchscreen pumps, managing the complete software development lifecycle from initial design through production release (Python, Go, Vue.js, TypeScript)
- Spearheaded development of the Masterflex Inventory Management System as a serverless application on AWS, utilizing CloudFormation, Cognito, API Gateway, Lambda, S3, and DynamoDB to collect, store, and analyze production and quality metrics
- Drove key contributions to Tangential Flow Filtration monitoring and analysis web application, encompassing software architecture design, implementation, and release (C++, Flutter)
- Optimized EOL tester application by enhancing application performance and refining test cases to efficiently identify hardware failures, yielding measurable improvements in product quality and reduced manufacturing defects
- Facilitated the successful launch of twelve new pump models by conducting thorough research and testing of new features to develop comprehensive test suites for EOL validation
- Automated manual system and smoke testing processes using ROBOT Framework and selenium, greatly increasing testing efficiency and reliability
- Engineered the serial communications module for embedded pump software, enabling RS-232 protocol-based pump control and monitoring capabilities
- Successfully transitioned responsibilities from external vendor Planet Innovation to in-house operations, demonstrating technical ownership and generating \$500,000+ in annual cost savings
- Collaborated with cross-functional teams including software engineers, firmware engineers, mechanical engineers, and end-users to drive continuous software improvements and maintain effective project communication

Software Engineering Intern at iotaMotion

Jun 2020 to Feb 2021

Iowa City, IA

- Researched and developed a sophisticated web application utilizing Django and PostgreSQL to store, analyze, and present auditory health data, which played a role in supporting the company's research initiatives and clinical trials.
- Demonstrated strong problem-solving skills by designing and implementing signal processing algorithms, enabling seamless processing and visualization of electrocochleography data
- Worked closely with engineers and engaged potential end-users to gather feedback, leading to improvements that tailored the application to the users' needs
- Began development on a computer vision based mobile application using React Native to facilitate the collection and storage of auditory health data

Leadership Experience

Teaching Assistant – Engineering Fundamentals II: Circuits

Aug 2019 to Aug 2021

- Led 50-minute discussion classes each week to supplement topics covered in lecture and work through problems
- Held office hours for 5 hours each week to assist students with homework and answer course-related questions
- Led review sessions for up to 60 students before midterms and finals to help students prepare and answer questions

Skills

Python, C++, Java, JavaScript, Golang, MATLAB, HTML, CSS, TypeScript, Linux, PostgreSQL, NoSQL, DynamoDB, Django, Flask, Asynchronous Programming, Vue.js, Node.js, React.js, Flutter, Git, Docker, AWS, Serverless Architecture, Full-stack Web Development, OOP, Agile, Artificial Intelligence, Computer Vision, Signal Processing, Image Processing, Embedded Linux, Unit Testing, Integration Testing, Automation Testing